

Quality Assurance

Software quality fundamentals, defect prevention and QA across the lifecycle.

What is Quality Assurance?

Quality Assurance (QA) is a systematic approach to increasing confidence in product and process quality. QA works before, during and after test execution by reviewing requirements, identifying risks, improving processes, promoting testability and providing evidence for decisions.

QA, QC and Testing

QA is prevention and process oriented. Quality Control (QC) observes the product and its outcomes. Testing is a verification and validation activity that reveals information about behavior, risks and defects. Terminology may vary across organizations, so context matters.

Error, defect and failure

An error is an incorrect human action; a defect is an imperfection introduced into an artifact; a failure is incorrect behavior observed when a defect manifests. Example: interpreting 15% as 5% is an error; implementing 5% is the defect; the customer receiving 5% is the observed failure.

Verification and validation

Verification asks whether we are building the product according to specifications and requirements. Validation asks whether we are building the right product for the real need. QA uses both to prevent a technically correct solution from solving the wrong problem.

Quality across the lifecycle

QA can contribute to refinement, acceptance criteria, design, development, testing, release and production. Earlier risk discovery usually gives teams more options to correct problems with lower impact. This connects to Shift Left, Quality by Design and Whole Team Quality.

Quality Engineering

Quality Engineering extends quality into architecture, automation, CI/CD, observability, testability, data and production. The goal is not only to find bugs but to build mechanisms that make quality continuously assessable.

What does QA do day to day?

QA analyzes requirements, identifies risks, designs scenarios, prepares data, executes tests, investigates logs and APIs, reports defects, retests fixes, runs regression, follows metrics and collaborates with engineering, product and other stakeholders.

Common beginner mistakes

Treating QA as test-case execution only; testing only happy paths; reporting bugs without evidence; ignoring risk and context; assuming 100% executed cases means 100% product coverage; and automating before understanding what behavior must be validated.

QA tip

Quality belongs to the whole team; QA helps make risks and evidence visible.